



SALES FORECASTING USING SARIMA



🔥 Project Overview

This project focuses on Sales Forecasting using the SARIMA (Seasonal Auto Regressive Integrated Moving Average) model.

The objective is to analyze historical sales data, identify trend and seasonal patterns, and forecast future sales using time-series analysis.

The project follows an end-to-end forecasting workflow covering:

Data preprocessing

Exploratory Data Analysis

Time-series analysis

Stationarity analysis

SARIMA model development

Forecast generation

Model evaluation

Business insights

Business Objective

Accurate sales forecasting can help organizations make better decisions related to:

 Inventory management

 Revenue planning

 Demand forecasting

 Sales strategy

 Resource allocation

 Business planning

The goal of this project is to use historical sales patterns to generate reliable future sales forecasts.

Technologies & Tools

Category	Tools
Programming	Python
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib
Statistical Modeling	SARIMA
Statistical Analysis	Statsmodels
Development Environment	Google Colab / Jupyter Notebook

Project Workflow

Historical Sales Dataset



Data Loading



Data Cleaning & Preprocessing



Exploratory Data Analysis



Time-Series Visualization



Stationarity Analysis



Train-Test Split



SARIMA Modeling



Forecasting



Model Evaluation



Business Insights

Project Steps

1. Data Loading

The historical sales dataset was imported using Pandas and analyzed to understand the available variables, data types, and overall structure.

2. Data Cleaning & Preprocessing

The dataset was prepared for time-series forecasting by:

Handling missing values

Converting date variables into datetime format

Sorting the dataset chronologically

Checking data consistency

Preparing sales data for time-series analysis

3. Exploratory Data Analysis

Historical sales were visualized to identify:

Overall sales trends

Seasonal patterns

Sales fluctuations

Growth and decline periods

Potential anomalies

4. Stationarity Analysis

Stationarity was examined because SARIMA requires the underlying time series to be appropriately transformed.

Differencing was applied where necessary to reduce non-stationarity and make the series suitable for modeling.

SARIMA Model

SARIMA (Seasonal AutoRegressive Integrated Moving Average) is an extension of the ARIMA model designed to handle seasonal time-series patterns.

SARIMA is represented as:

$SARIMA(p, d, q)(P, D, Q, s)$

Where

$p \rightarrow$ AutoRegressive order

$d \rightarrow$ Degree of differencing

$q \rightarrow$ Moving Average order

$P \rightarrow$ Seasonal AutoRegressive order

$D \rightarrow$ Seasonal differencing

$Q \rightarrow$ Seasonal Moving Average order

$s \rightarrow$ Seasonal period

The model was used to capture both non-seasonal and seasonal dependencies within the historical sales data.

Model Evaluation

The SARIMA forecasting model was evaluated using:

Metric	Result
MAE	13,930.02
RMSE	16,394.82
MAPE	27.77%

Evaluation Metrics

MAE — Mean Absolute Error

Measures the average absolute difference between actual and predicted sales.

RMSE — Root Mean Squared Error

Measures prediction error while giving greater importance to larger errors.

MAPE — Mean Absolute Percentage Error

Measures the average percentage difference between actual and predicted sales.

Forecast Analysis

The project compares historical sales with SARIMA-generated forecasts to evaluate how effectively the model captures the underlying sales patterns.

The forecast visualization helps identify:

Historical sales movement

Predicted sales

Seasonal behavior

Forecast trends

Differences between actual and predicted values

Key Insights

The analysis demonstrates how historical sales patterns can be used to forecast future demand.

Key Takeaways

Historical sales data contains identifiable time-dependent patterns.

Seasonality plays an important role in sales forecasting.

SARIMA can model both trend-related and seasonal behavior.

Forecasting can support inventory and revenue planning.

Evaluation metrics provide an objective measure of model performance.

Visualization makes forecast results easier to communicate to business stakeholders.

Business Applications

The forecasting approach can be applied to:

Retail sales forecasting

Inventory planning

Demand forecasting

Revenue estimation

Supply-chain planning

Financial planning

Resource allocation

Skills Demonstrated

Python

Pandas

NumPy

Matplotlib

Data Cleaning

Exploratory Data Analysis

Time-Series Analysis

Stationarity Analysis

SARIMA

Statistical Modeling

Forecasting

Model Evaluation

Business Insights

Project Structure

Sales-Forecasting/

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└─  Sales_Forecasting.ipynb

└─  train.csv

└─  images/

└─  README.md

Update the file names if your final GitHub repository uses different names.

Future Improvements

The project can be further enhanced by:

Hyperparameter optimization for SARIMA

ACF and PACF-based parameter selection

Time-series cross-validation

Comparison with alternative forecasting models

Incorporating external variables

Building an interactive Power BI forecasting dashboard

Deploying the forecasting model as a web application

SOCIAL LINKS

 GitHub: <https://github.com/skm-1718>

 LinkedIn: <https://www.linkedin.com/in/sangram-keshari-mohapatra-9104411b/>

Project Highlight

An end-to-end Sales Forecasting project using SARIMA to analyze historical sales patterns, capture seasonality, and generate future sales predictions for data-driven business planning.